Indian Institute of Technology
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Mid-Way Evaluation Report

Personalised Mental Health Digital Twin using AI

Problem Statement: Personalised Mental Health Digital Twin using AI

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1 Introduction

This project aims to develop a personalised mental health digital twin that monitors behavioural changes over time through journal entries. Instead of comparing users against fixed thresholds, the system builds an individual baseline, detects anomalies, and identifies possible mental health concerns using an explainable PST-based risk assessment framework. The current implementation focuses on baseline creation, anomaly detection, risk assessment, frontend design, and the overall system architecture.

2 Problem Understanding

2.1 Problem Statement

The Problem Statement requires us to create a system that can track an individual's mental state over time and warn the user of risks and anomalies identified by comparing the user's normal behaviour with the behaviour currently observed.

2.2 Objectives

- Develop a system that extracts psychological parameters from journal entries using an LLM.
- Build personalised behavioural baselines for each user to monitor changes over time.
- Detect significant and persistent deviations from a user's normal behavioural patterns.
- Assess the risk of mental health issues through an explainable, parameter-based framework.
- Present behavioural trends and risk insights in an interpretable manner to assist early intervention.

3 Functional Requirements

3.1 Core Features

- **Journal Entry Management -** Allow users to submit and store journal entries for analysis.
- **LLM-based Parameter Extraction -** Extract psychological parameters from journal entries using a Hugging Face transformer model.
- **Personalised Baseline Creation -** Build and continuously update a behavioural baseline for each user.
- **Anomaly Detection -** Detect significant deviations from the user's personalised baseline.
- **Risk Assessment -** Identify potential mental health issues using the PST-based explainable risk assessment framework.
- **Behavioural Trend Visualisation -** Display parameter trends and risk progression over time.

3.2 Additional Features

- **Explainable Insights -** Explain why a particular risk warning was generated by highlighting the contributing parameters.
- **Historical Progress Tracking -** Allow users to review past journal entries and behavioural changes.
- **Persistent Anomaly Monitoring -** Reduce false positives by considering repeated anomalies over time instead of isolated deviations.
- **Personalised Recommendations (Future Scope) -** Suggest possible actions or coping strategies based on detected behavioural changes.

4 System Components

Module	Technology	Purpose
Frontend	HTML, CSS	Provides the user interface for journal entry submission and displays behavioural trends and risk assessment outputs.
Parameter Extraction	Python, Hugging Face Transformers	Analyses journal entries using an LLM and extracts numerical scores for the selected psychological parameters.
Data Storage & Personalised Baseline	Python, SQLite, NumPy, Pandas	Stores user data and journal entries, and computes personalised behavioural baselines using weighted averaging and outlier dampening.
Anomaly Detection	Python, PatchTST	Learns the user's normal behavioural patterns, detects significant deviations, and identifies the parameters contributing to anomalies.
Risk Assessment	Python	Analyses anomalous parameters using the PST framework to identify potential mental health issues and generate explainable warnings.
Version Control	Git, GitHub	Supports collaborative development and version management.

4.1 Parameter Extraction

The parameter extraction module converts unstructured journal entries into 22 normalised psychological parameter scores using a Hugging Face Transformer model. These scores represent the user's behavioural and emotional state and serve as input to the baseline, anomaly detection, and risk assessment modules.

4.2 Data Storage and Personalised Baseline

The personalised baseline is maintained separately for each user instead of relying on universal thresholds. The baseline is updated using an exponentially decaying weighted average,

$$w = e^{-0.1\times \text{days\_ago}}$$

where recent entries contribute more than older ones. Outlier dampening reduces the influence of extreme observations so the baseline can adapt gradually without being distorted by short-term fluctuations.

4.3 Anomaly Detection

The anomaly detection module compares incoming parameter scores with the user's historical behavioural pattern. A PatchTST time-series model predicts expected parameter values, and significant deviations are flagged as anomalies. The module also identifies the most influential anomalous parameters and computes a baseline update weight.

4.4 Risk Assessment

The risk assessment module evaluates anomalous parameters using a Primary–Secondary–Tertiary (PST) evidence framework developed from psychological assessment scales and mental health literature. For each issue, Primary, Secondary, and Tertiary indicators are assigned weights of 3, 2, and 1 respectively. The weighted risk score is calculated as:

$$\text{Risk Score} = 3(\text{Primary}\%) + 2(\text{Secondary}\%) + 1(\text{Tertiary}\%)$$

The output remains explainable by reporting anomalous Primary indicators as Core Indicators and anomalous Secondary indicators as Supporting Indicators.

5 System Architecture

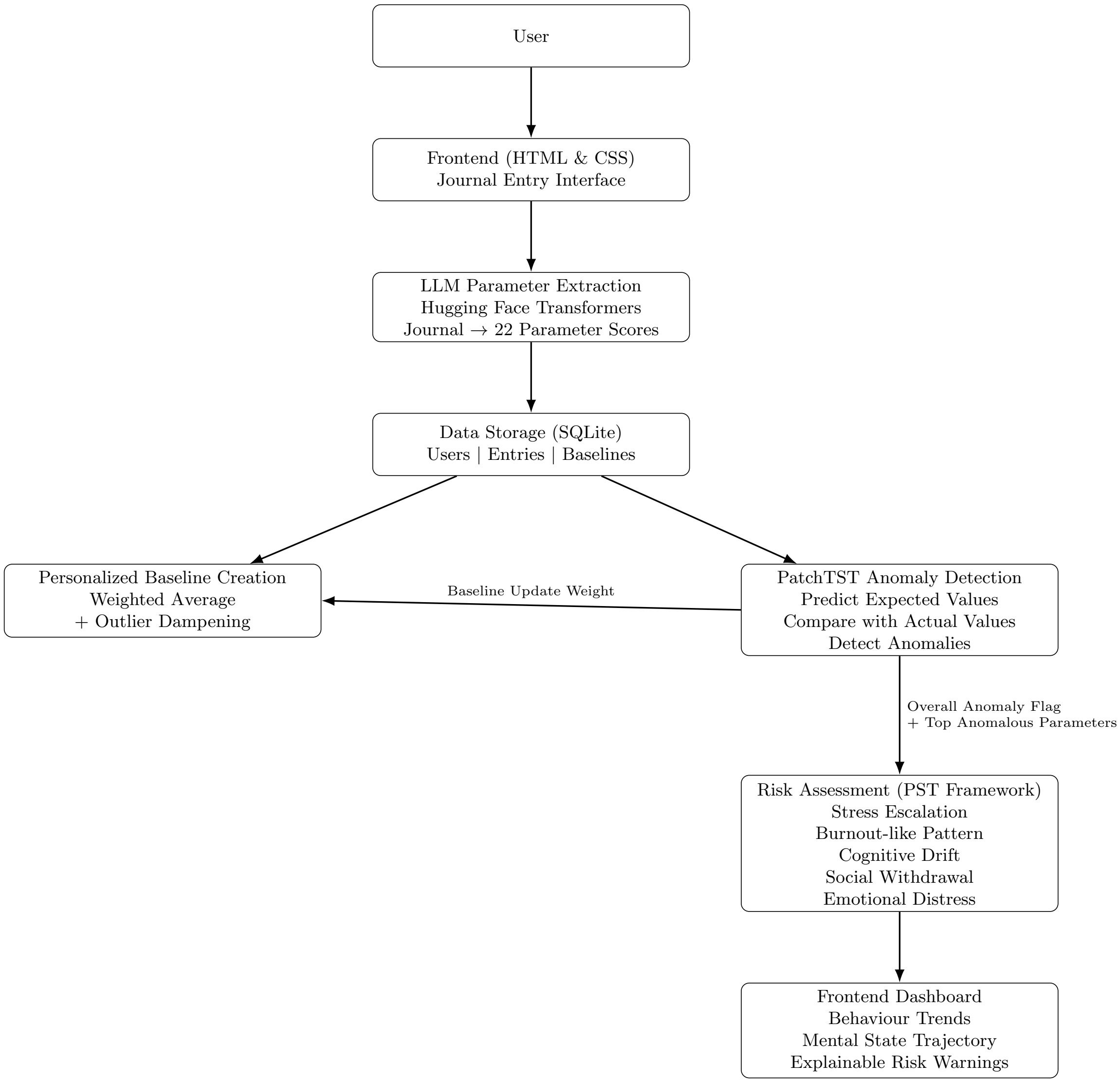


Figure 1: System Architecture of the Mental Health Digital Twin

6 Database Design

The system uses SQLite to store user information, journal entries, extracted psychological parameters, and personalised baselines. The Users, Entries, and Baselines tables support retrieval of historical data required for baseline creation, anomaly detection, and risk assessment.

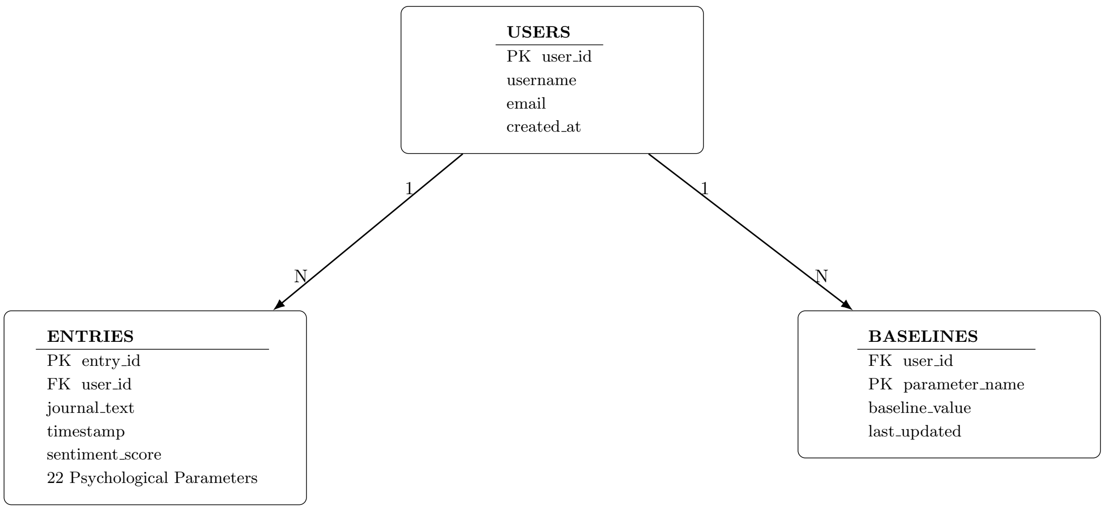


Figure 2: Entity Relationship (ER) Diagram

7 UI Design

The current UI is a basic HTML wireframe of the Digital Twin dashboard with a two-column layout. The left side is designed for daily journal input, while the right side is reserved for mental state trajectory visualisations and risk assessment outputs. Styling and backend integration are planned for later iterations.

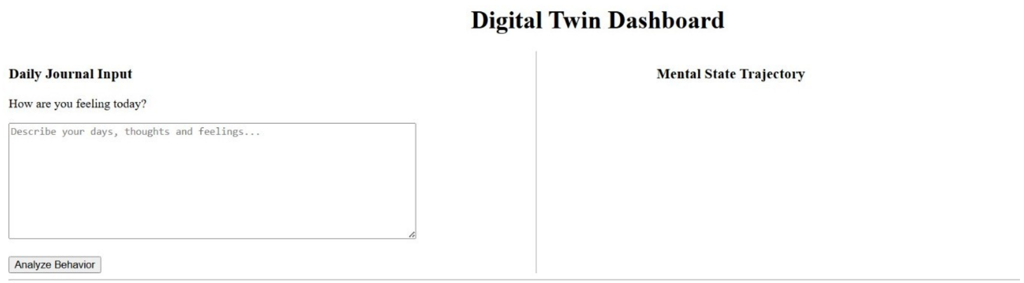


Figure 3: Initial HTML dashboard wireframe

8 Current Progress

8.1 Completed

- System architecture, database schema, and repository structure finalised.
- Data storage, personalised baseline, anomaly detection, and risk assessment modules implemented.
- Initial HTML frontend developed for journal input and visualisation.

8.2 In Progress

- Hugging Face-based parameter extraction and module integration.
- Anomaly reporting improvements, persistence analysis, and frontend refinement.

8.3 Yet to Start

- End-to-end integration, testing, and validation.
- Performance refinement, final documentation, and deployment preparation.

9 Future Work

Timeline	Planned Work
Week 1	Complete parameter extraction and integrate it with the baseline module.
Week 2	Integrate anomaly detection and implement PST-based risk assessment logic.
Week 3	Complete frontend integration, persistence analysis, and dashboard improvements.
Week 4	Perform testing, refinement, documentation, and final demonstration preparation.

10 GitHub Repository

Repository Link: https://github.com/Rayhann12345/Mental-Health-Digital-Twin

11 Implementation Progress

11.1 Data Storage and Personalised Baseline

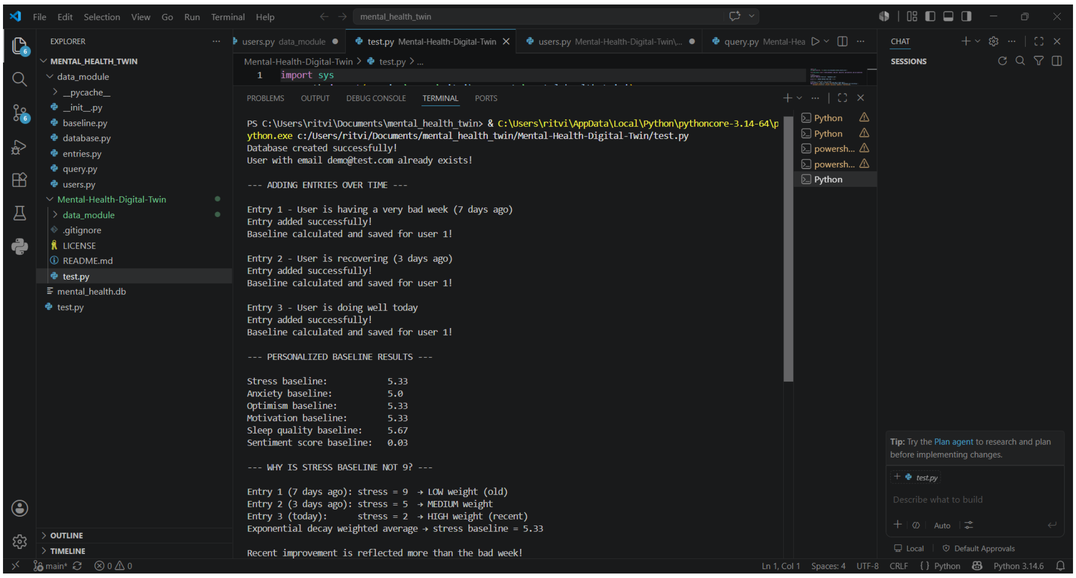
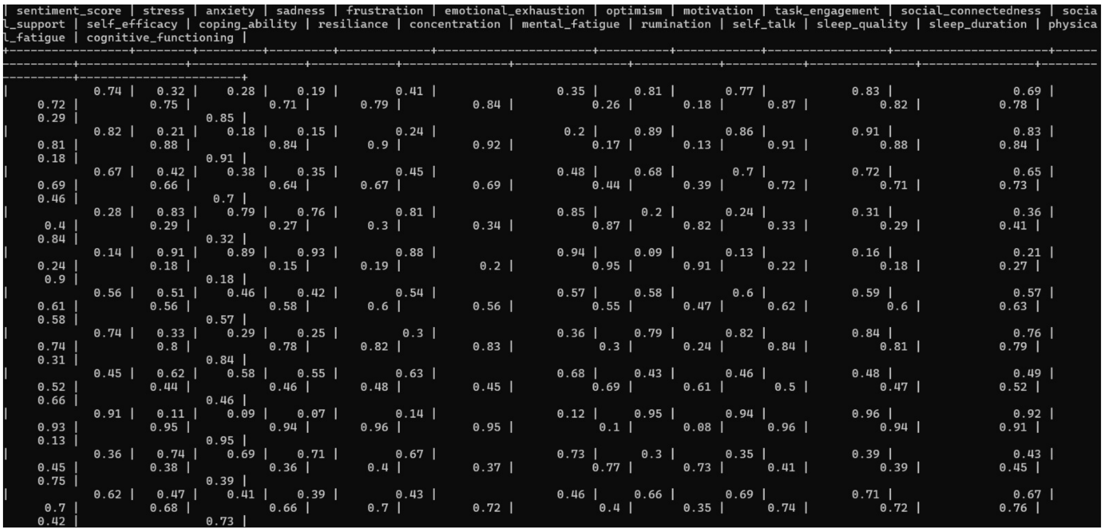


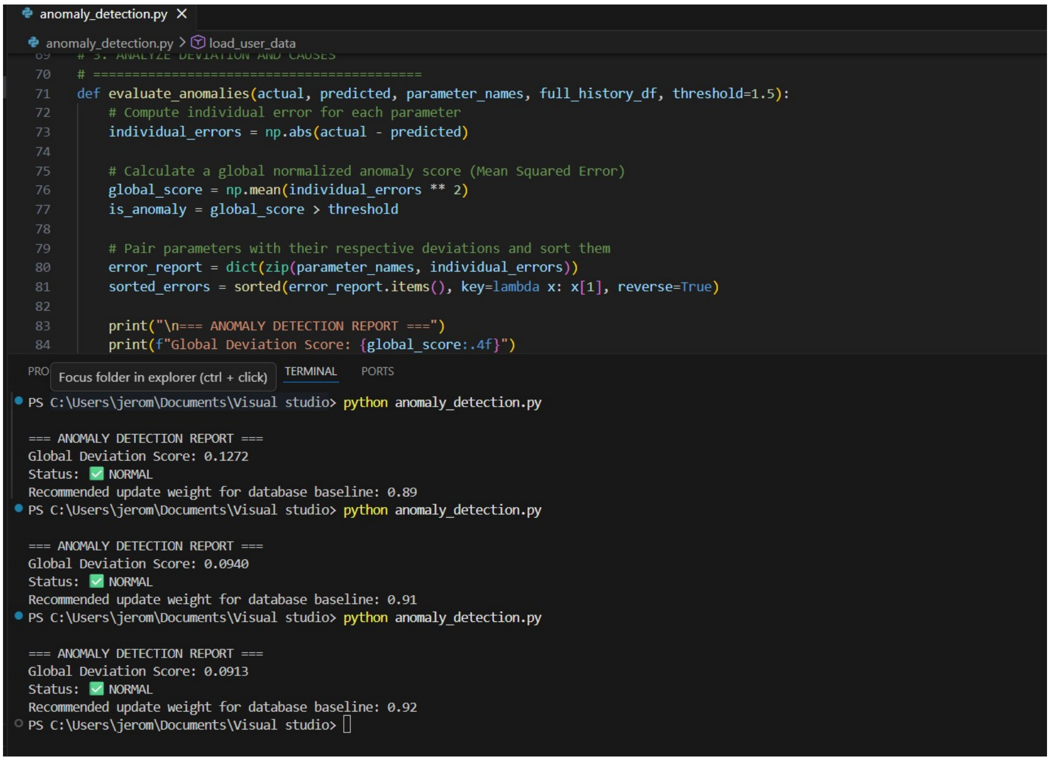
Figure 4: Terminal result showing stored entries and baseline calculation

11.2 Anomaly Detection



sentiment_score	stress	anxiety	sadness	frustration	emotional_exhaustion	optimism	motivation	task_engagement	social_connectedness	social_support	self_efficacy	coping_ability	resiliance	concentration	mental_fatigue	rumination	self_talk	sleep_quality	sleep_duration	physical_fatigue	cognitive_functioning
0.74	0.32	0.28	0.19	0.41	0.35	0.81	0.77	0.83	0.69	0.72	0.75	0.71	0.79	0.84	0.26	0.18	0.87	0.82	0.78	0.29	0.85
0.82	0.21	0.18	0.15	0.24	0.2	0.89	0.86	0.91	0.83	0.81	0.88	0.84	0.9	0.92	0.17	0.13	0.91	0.88	0.84	0.18	0.91
0.67	0.42	0.38	0.35	0.45	0.48	0.68	0.7	0.72	0.65	0.69	0.66	0.64	0.67	0.69	0.44	0.39	0.72	0.71	0.73	0.46	0.7
0.28	0.83	0.79	0.76	0.81	0.85	0.2	0.24	0.31	0.36	0.4	0.29	0.27	0.3	0.34	0.87	0.82	0.33	0.29	0.41	0.84	0.32
0.14	0.91	0.89	0.93	0.88	0.94	0.09	0.13	0.16	0.21	0.24	0.18	0.15	0.19	0.2	0.95	0.91	0.22	0.18	0.27	0.9	0.18
0.56	0.51	0.46	0.42	0.54	0.57	0.58	0.6	0.59	0.57	0.61	0.56	0.58	0.6	0.56	0.55	0.47	0.62	0.6	0.63	0.58	0.57
0.74	0.33	0.29	0.25	0.3	0.36	0.79	0.82	0.84	0.76	0.74	0.8	0.78	0.82	0.83	0.3	0.24	0.84	0.81	0.79	0.31	0.84
0.45	0.62	0.58	0.55	0.63	0.68	0.43	0.46	0.48	0.49	0.52	0.44	0.46	0.48	0.45	0.69	0.61	0.5	0.47	0.52	0.66	0.46
0.91	0.11	0.09	0.07	0.14	0.12	0.95	0.94	0.96	0.92	0.93	0.95	0.94	0.96	0.95	0.1	0.08	0.96	0.94	0.91	0.13	0.95
0.36	0.74	0.69	0.71	0.67	0.73	0.3	0.35	0.39	0.43	0.45	0.38	0.36	0.4	0.37	0.77	0.73	0.41	0.39	0.45	0.75	0.39
0.62	0.47	0.41	0.39	0.43	0.46	0.66	0.69	0.71	0.67	0.7	0.68	0.66	0.7	0.72	0.4	0.35	0.74	0.72	0.76	0.42	0.73

Figure 5: Custom sample dataset used for testing the anomaly detection framework



anomaly_detection.py ×

anomaly_detection.py > load_user_data

```python
# 3. ANALYZE DEVIATION AND CAUSES
# ==========================================
def evaluate_anomalies(actual, predicted, parameter_names, full_history_df, threshold=1.5):
    # Compute individual error for each parameter
    individual_errors = np.abs(actual - predicted)

    # Calculate a global normalized anomaly score (Mean Squared Error)
    global_score = np.mean(individual_errors ** 2)
    is_anomaly = global_score > threshold

    # Pair parameters with their respective deviations and sort them
    error_report = dict(zip(parameter_names, individual_errors))
    sorted_errors = sorted(error_report.items(), key=lambda x: x[1], reverse=True)

    print("\n=== ANOMALY DETECTION REPORT ===")
    print(f"Global Deviation Score: {global_score:.4f}")
```

PRO Focus folder in explorer (ctrl + click) TERMINAL PORTS

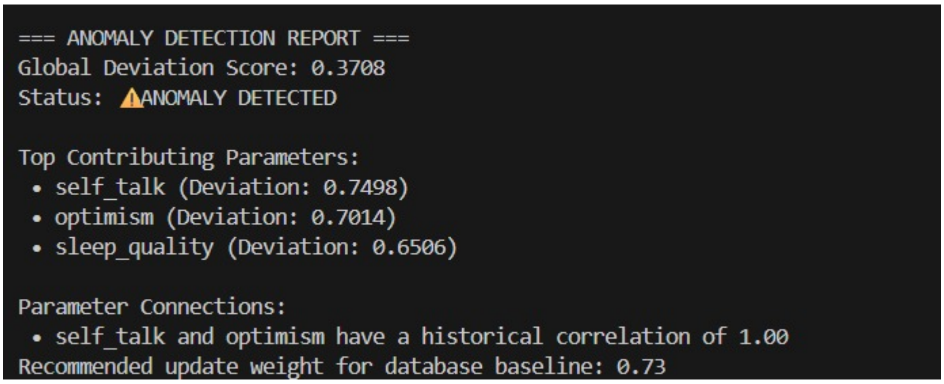
```
PS C:\Users\jerom\Documents\Visual studio> python anomaly_detection.py

=== ANOMALY DETECTION REPORT ===
Global Deviation Score: 0.1272
Status: ✅ NORMAL
Recommended update weight for database baseline: 0.89
PS C:\Users\jerom\Documents\Visual studio> python anomaly_detection.py

=== ANOMALY DETECTION REPORT ===
Global Deviation Score: 0.0940
Status: ✅ NORMAL
Recommended update weight for database baseline: 0.91
PS C:\Users\jerom\Documents\Visual studio> python anomaly_detection.py

=== ANOMALY DETECTION REPORT ===
Global Deviation Score: 0.0913
Status: ✅ NORMAL
Recommended update weight for database baseline: 0.92
PS C:\Users\jerom\Documents\Visual studio>
```

Figure 6: Terminal output for anomaly detection testing



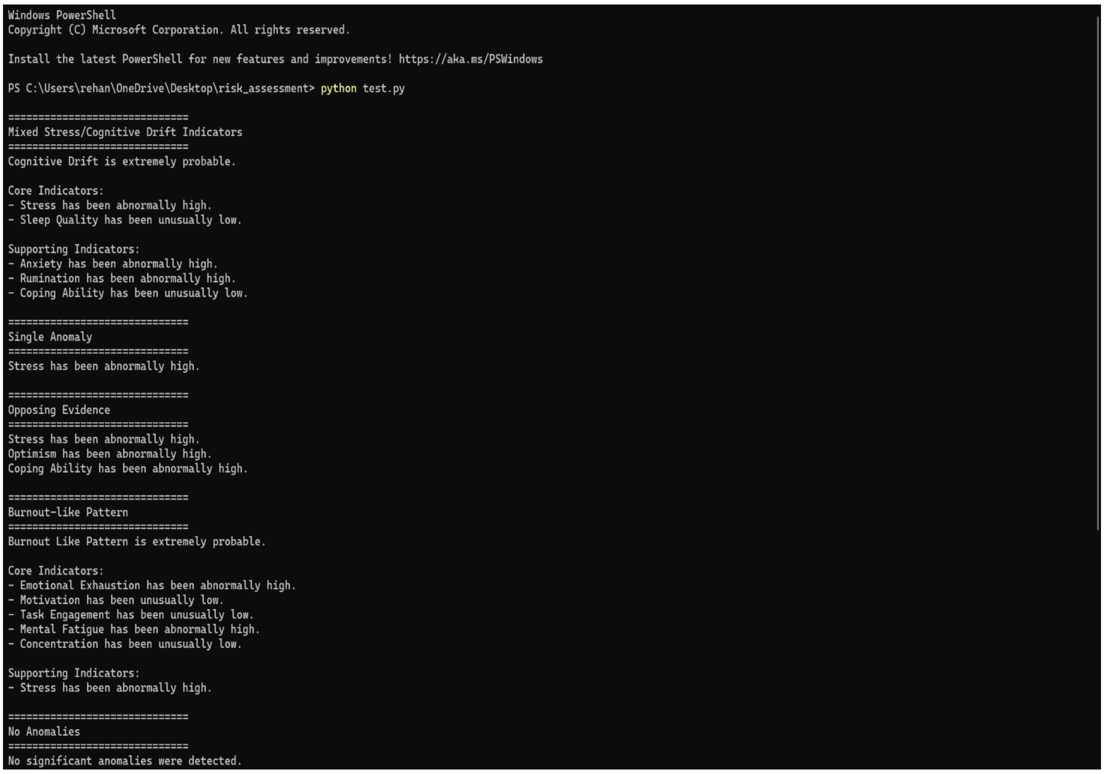
```
=== ANOMALY DETECTION REPORT ===
Global Deviation Score: 0.3708
Status: ⚠ANOMALY DETECTED

Top Contributing Parameters:
 • self_talk (Deviation: 0.7498)
 • optimism (Deviation: 0.7014)
 • sleep_quality (Deviation: 0.6506)

Parameter Connections:
 • self_talk and optimism have a historical correlation of 1.00
Recommended update weight for database baseline: 0.73
```

Figure 7: Terminal output showing anomaly detected and contributing parameters

11.3 Risk Assessment


```
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows

PS C:\Users\rehan\OneDrive\Desktop\risk_assessment> python test.py

==============================
Mixed Stress/Cognitive Drift Indicators
==============================
Cognitive Drift is extremely probable.

Core Indicators:
- Stress has been abnormally high.
- Sleep Quality has been unusually low.

Supporting Indicators:
- Anxiety has been abnormally high.
- Rumination has been abnormally high.
- Coping Ability has been unusually low.

==============================
Single Anomaly
==============================
Stress has been abnormally high.

==============================
Opposing Evidence
==============================
Stress has been abnormally high.
Optimism has been abnormally high.
Coping Ability has been abnormally high.

==============================
Burnout-like Pattern
==============================
Burnout Like Pattern is extremely probable.

Core Indicators:
- Emotional Exhaustion has been abnormally high.
- Motivation has been unusually low.
- Task Engagement has been unusually low.
- Mental Fatigue has been abnormally high.
- Concentration has been unusually low.

Supporting Indicators:
- Stress has been abnormally high.

==============================
No Anomalies
==============================
No significant anomalies were detected.
```

Figure 8: Terminal output for risk assessment testing

12 References

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